Shivam Kumar

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Education

Indian Institute of Technology, Kanpur — CGPA: 9.4/10.0
Bachelor of Technology in Computer Science and Engineering — *2017 – 2021*

Sunbeam English School, CBSE — 96.4%
AISSCE (Class XII) — *2017*

Pristine Children's High School — CGPA: 10/10
AISSE (Class X) — *2015*

Experience

Software Development Engineer II — Jul 2024 – Present
Microsoft – Azure Container Apps — *Hyderabad, India*

- Designed and deployed **AI-powered SRE agents** for Azure Container Apps incident diagnostics, reducing P50 time-to-mitigate by **97% (185→6 min)** and time-to-resolve by **64%**; evolving toward **autonomous auto-repair** with automated RCA comparison; enabled **Foundry Control Plane** for ACA-hosted agents.
- Built **Session Management APIs** and fixed critical async execution bugs for ACA Sessions, improving stability for **∼300 external customers**; enabled early session deletion for the Fabric team, optimizing billing and costs.
- Expanded **Azure Functions on ACA** platform — function keys management, invocation tracing, durable-events logging, KEDA scaling rules, and drain functionality, benefiting **∼1,322 external apps**.
- Led **Dapr AKS Extension** rollouts (v1.15, v1.16) across all Azure regions; remediated CVEs across **∼400 AKS clusters**; integrated CodeQL for pipeline security.
- Integrated **Azure Key Vault** with Managed Environment Storage API, securing **∼27K daily API calls** by eliminating raw connection strings.
- Served as **on-call DRI** resolving **200+ incidents** including 27+ CRIs, outperforming team baselines.

Software Development Engineer — Jul 2021 – Jul 2024
Microsoft – Azure Container Apps / Dapr — *Hyderabad, India*

- Contributed to **Dapr** (CNCF open-source project) — improved several components and APIs to stable; implemented **cascade terminate/purge** for Dapr Workflows across runtime and SDKs; added bulk subscribe support in Azure Event Hubs; added workflow performance tests identifying P0 bugs.
- Enabled **airgap installation**, winget, and MSI for Dapr CLI; published combined coverage reports; resolved customer issues on GitHub and Discord.
- Architected **internal fork and CI/CD infrastructure** across 5 Dapr repositories; automated OSS-to-internal releases within **<1 day SLA**; consolidated into **OneBranch-compliant** pipelines with Ev2 deployment.
- Added **multi-arch support** (ARM64, ARM, AMD64), Mariner and distroless images; handled 1P Container Registry migration; delivered nightly builds for ACA; contributed to AKS Extension GA requirements.
- Helped launch **ACA Sessions public preview** — added support for Node.js code interpreter sessions; fixed critical bugs in Python and Node.js sessions; added performance tests identifying regressions.

Quantitative Research Intern — May 2020 – Jul 2020
JP Morgan & Chase – E-Trading — *Mumbai, India (Remote)*

- Built price prediction models using **LSTM**, feedforward neural networks, and regression with univariate feature selection.
- Improved model predictive power by **40%** for HK names through feature standardization and data preprocessing.
- Implemented L1 and L2 features for optimizing model parameterization across multiple asset classes.

Research

Data Race Detection for Task-Parallel Programs — Jun 2019 – Aug 2021
Supervisor: Prof. Swarnendu Biswas, IIT Kanpur

- Designed novel algorithm **Tasker** integrating vector clocks with tree-based techniques; implemented using **LLVM** pass for memory instrumentation on task-parallel programs.
- Achieved **1.48X speedup** over PTRacer baseline on 128GB Intel Xeon system with lower memory overhead.
- *"Efficient Data Race Detection of Structured Task Parallel Programs Using Vector Clocks"* — S. Kumar, A. Agrawal, S. Biswas | Submitted to ISCGO 2022.

Projects

Java Compiler | *Python, PLY, Compiler Design (CS335)* — Jan 2020 – Apr 2020
- Designed lexer, parser, and 3-address code generation; support for symbol tables, classes, functions, and interfaces.

GemOS | *C, x86, Operating Systems (CS330)* — Aug 2019 – Nov 2019
- Implemented mmap, munmap, mprotect with lazy allocation and pagefault handling; cfork/vfork with copy-on-write.

Machine Learning | *Python, Keras, ML (CS771)* — Aug 2019 – Nov 2019
- CNN image classification using Keras; SGD for binary classification; multi-label Bonsai recommender system.

Cipher Decoder | *Python, Modern Cryptology (CS641)* — Aug 2020 – Nov 2020
- Differential cryptanalysis of 3-DES; decryption of Caesar, Vigenere ciphers using frequency analysis.

SAT Solver | *Python, Logic in CS (CS202)* — Jan 2019 – Apr 2019
- DPLL-based SAT solver; encoded diagonal sudoku constraints in DIMACS propositional logic format.

MERN Mobile App | *React Native, Node.js, MongoDB (CS252)* — May 2019 – Jul 2019
- Full-stack mobile app with React Native frontend, Node.js/Express backend, MongoDB, and secure JWT login.

Technical Skills

Languages: Go, Python, C#/.NET, JavaScript/TypeScript, C/C++
Cloud & Infrastructure: Azure (Container Apps, AKS, Functions, Key Vault, Event Hubs, CosmosDB), Kubernetes, Docker, Dapr (CNCF), Helm
Tools & CI/CD: OneBranch (1ES), Ev2, Azure DevOps Pipelines, GitHub Actions, Git, Geneva, CodeQL, KEDA
Domains: Distributed Systems, AI/LLM Agents, CI/CD & Release Engineering, Microservices, Site Reliability (SRE)

Achievements & Leadership

RockStar Award, Microsoft ('22, '23) for exceptional contributions | **Academic Excellence Award** (×2), IIT Kanpur
JEE Advanced **AIR 715** | JEE Mains **AIR 348** | KVPY Scholar | NSE Physics **Top 1%** | Ram Prakash Chopra Scholarship
Academic Mentor, IIT Kanpur ('18-19) | Secretary, Dramatics Club, IIT Kanpur ('18-19)